

Log Investigation

Objective :

To simulate a brute-force SSH login attack and analyze authentication logs to identify suspicious login activity.

Tools Used :

- Kali Linux
- OpenSSH Server
- journalctl
- grep
- Terminal

Commands Used :

```
ssh kali@10.0.2.15
```

```
journalctl | grep ssh
```

```
journalctl | grep "Failed"
```

```
journalctl | grep password
```

Steps Performed :

1. Started the SSH service on Kali Linux.
2. Identified the local IP address of the system.
3. Attempted SSH login using the command:
`ssh kali@10.0.2.15`
4. Entered incorrect passwords multiple times to simulate brute-force behavior.
5. Investigated authentication logs using journalctl and grep commands.
6. Analyzed repeated failed login attempts from the same IP address.

Observations :

- Multiple failed SSH authentication attempts were recorded.
- Repeated login failures targeted the kali user account.
- Authentication events were successfully stored in system logs.

Findings :

- Suspicious repeated login attempts were identified.
- Logs indicated possible brute-force attack behavior.
- SSH authentication failures originated from IP address 10.0.2.15.
- Multiple failed password attempts occurred within a short time period.

Conclusion :

The investigation successfully demonstrated how brute-force attacks generate authentication logs that can be monitored and analyzed by SOC analysts. Repeated failed login attempts are a strong indicator of unauthorized access attempts.

Recommendations :

- Enable account lockout policies.
- Restrict SSH access to trusted IP addresses.
- Use strong passwords and multi-factor authentication.
- Continuously monitor authentication logs.
- Implement SIEM alert rules for repeated failed login attempts.

SOC Relevance :

This project demonstrates core SOC analyst activities including:

- Log monitoring
- Authentication analysis
- Threat detection
- SSH attack investigation
- Incident identification

Screenshots :

[illegible]